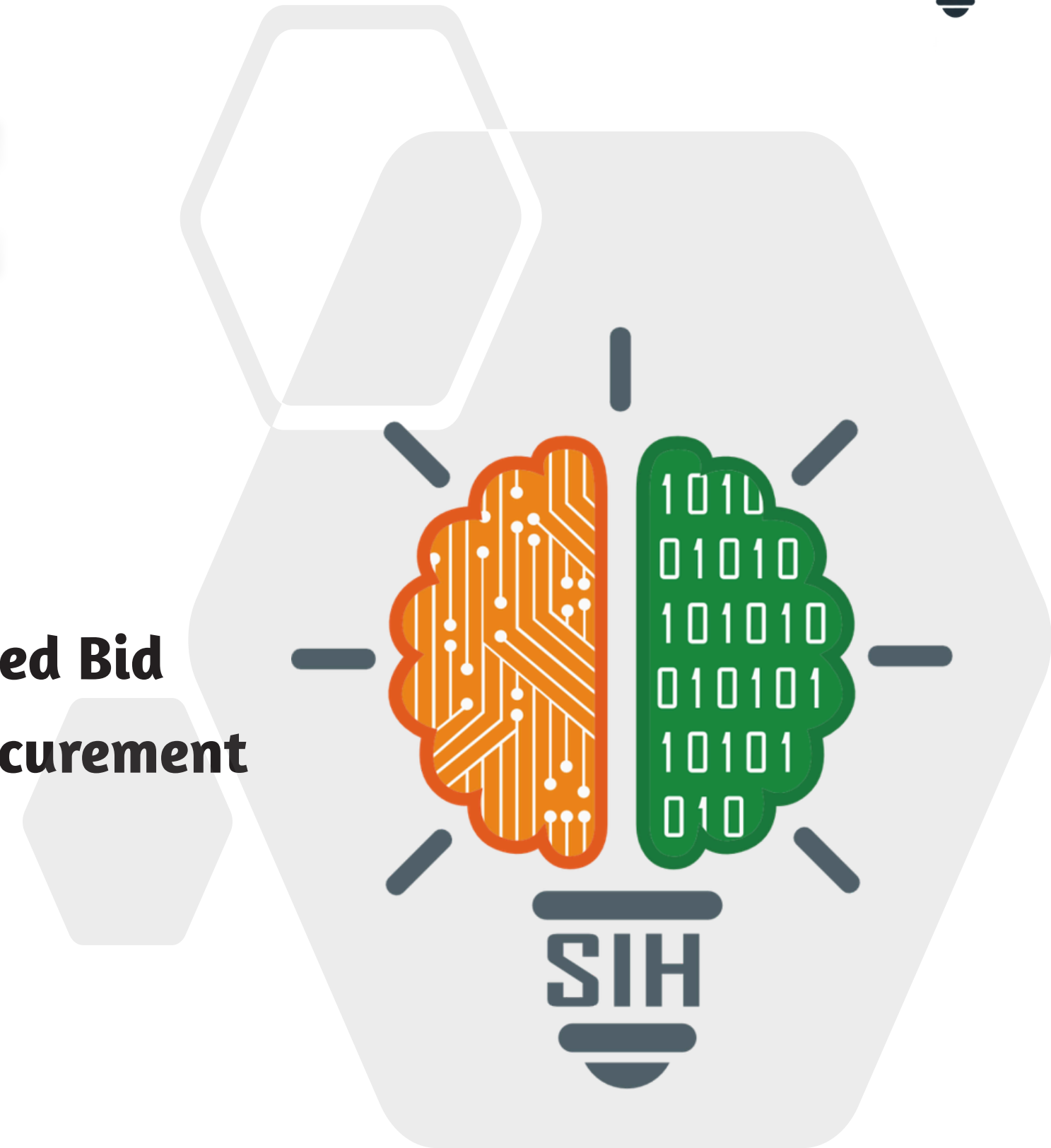


SMART INDIA HACKATHON 2026

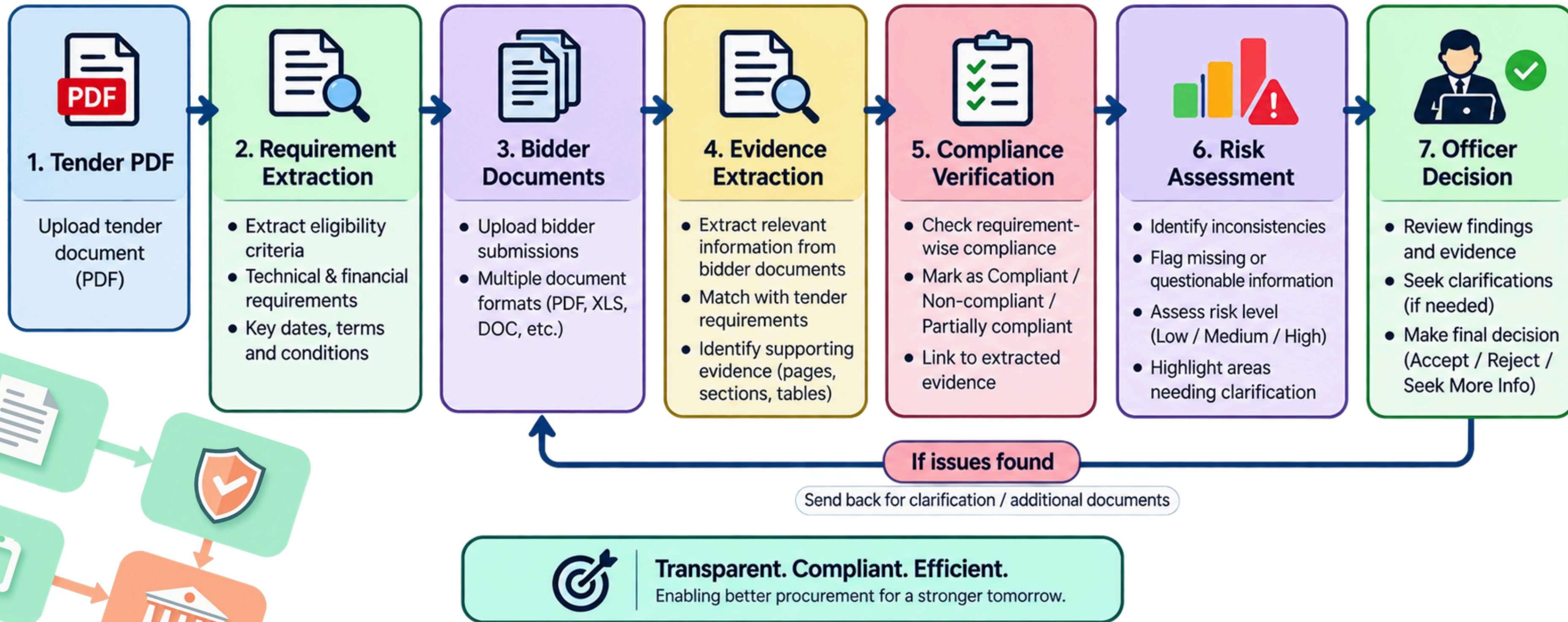


VERICORE

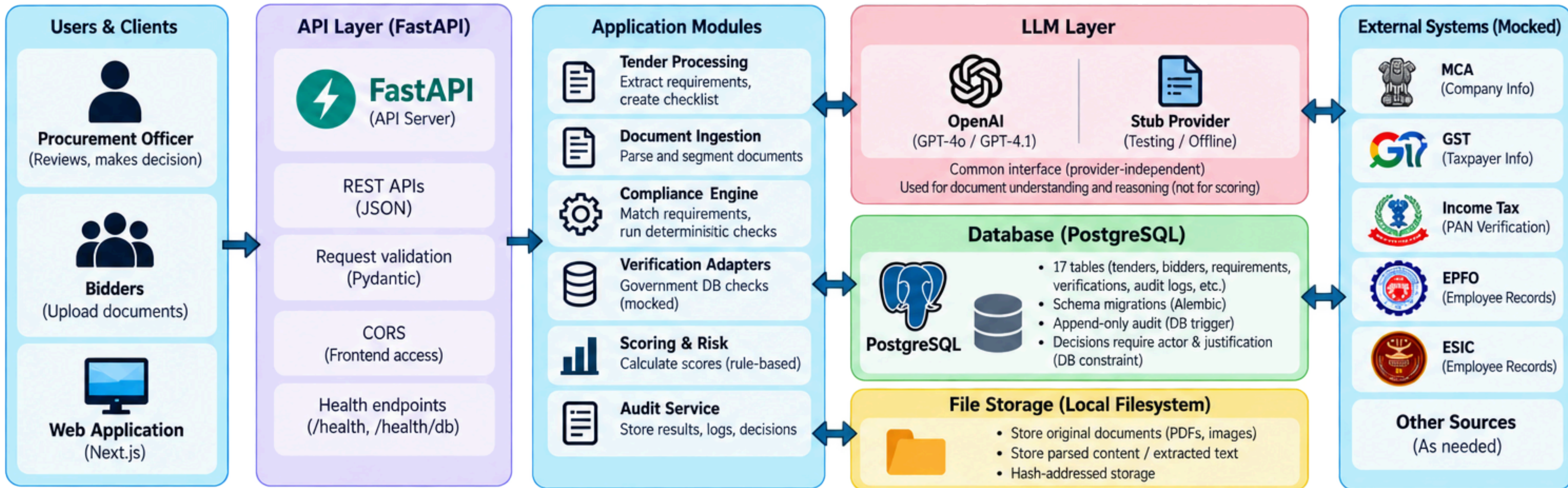
- **Problem Statement ID:** 26100
- **Problem Statement Title:** AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:**
- **Team Name:** VERICORE



OUR PROPOSED SOLUTION



Backend Architecture & Technical Stack



Technology Stack



Python 3.11+



FastAPI



PostgreSQL

Alembic
(migrations)Docker
(containerization)Git
(version control)VS Code
(development)

Technical Feasibility



Modular Technology Stack

Uses proven technologies like Next.js, FastAPI, PostgreSQL and PyMuPDF that can be integrated efficiently.



AI + Rule-Based Verification

LLMs handle document understanding while deterministic rules perform critical compliance checks reliably.



Scalable Architecture

Separate frontend, backend, database and AI layers make the system easy to maintain and expand.

Potential Challenges



Poor-Quality Documents

Scanned or unclear PDFs can make accurate information extraction difficult.



Complex Tender Conditions

Different tenders may contain varying rules that require careful interpretation.



External Portal Dependency

Government portal availability or API limitations may affect automated verification.



Data Inconsistency Across Documents

Details may differ between submitted documents, requiring cross-document verification.

Mitigation Strategies



Human-in-the-Loop Review

Uncertain or unplaceable evidence is sent for human review instead of being automatically accepted.



Deterministic Compliance Rules

PAN, GSTIN, dates, turnover and other critical fields are checked using fixed validation rules.



Controlled AI Usage

LLM calls use caching, chunking and rate limiting, while simulated portal checks are clearly labelled.



IMPACTS & BENEFITS



IMPACTS

FOR GOVERNMENT PROCUREMENT



Reduced Manual Scrutiny



Scalable Tender Evaluation



Improved Verification Consistency



Earlier Risk Identification



Greater Procurement Transparency



Stronger Accountability



More Defensible Procurement Decisions



Modernized Compliance Workflow



BENEFITS

FOR OFFICERS & DEPARTMENTS



Automated Requirement Extraction



Requirement-Wise Verification



Evidence-Level Traceability



Explainable Compliance Checks



Human-in-the-Loop Decisions



Structured Reporting



Tamper-Evident Audit Trail



RESEARCH AND REFERENCES



GeM
Government
e Marketplace

Official government procurement platform used as the primary reference for understanding the GeM procurement workflow, bid structure, eligibility requirements, and tender documentation.

 <https://gem.gov.in/>



**GeM
BidPlus**

Used to study real-world GeM bid listings, including bid IDs, tender descriptions, submission deadlines, buyer information, and bid-related documents.

 <https://bidplus.gem.gov.in/all-bids>



TenderKart

Used as a secondary reference for aggregated tender information and understanding bidder participation, tender results, and procurement-market data across different sources.

 <https://tenderkart.in/>

